AUCKLAND INTERNATIONAL CAMPUS | FutureSkills | OTAGO POLYTECHNIC Te Kura Matatini ki Otago

# IA7621001 Automation and Robotics

## In-class practicals - Ballistic Behaviour SBA

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| Study Block: | Block 4, 2025: 13 October 2025 to 19 December 2025 |
| Course Leader: | Dr Senaka Amarakeerthi |
| Due date and time: | 10th November 2025 - 1.00PM to 3.00PM |
| Submission: | Moodle Submission and a demonstration. |
| Submission format: | Word document with all code snippets. |
| Weighting/Contribution: | 10% |
| Marks out of | 30 |
| Course Level | 7 |
| Credits | 15 |
| Learning outcomes covered: | LO 3 |
| Pass Criteria | Cumulative 50% pass |
| Version | 1.0 |

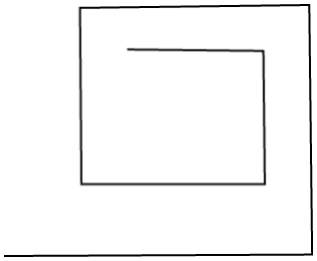
**Learning Outcomes Covered**
LO3: Use an appropriate software development platform to implement simple interactive robotics/automated systems.

**General Description and Conditions**

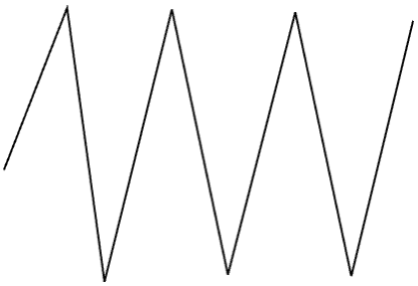
- This assessment task is designed to show you can apply core principles to robot design, design and implement a robotic solution to a problem
- The result of this task will accrue towards your final mark for this paper.
- SOFTWARE:
    - **Software must be your own work** and be uploaded to Moodle before the end of the session. In this early stage, I'm looking for performance from your robot rather than perfect code but try to make your code as neat and tidy as you can. Keep Loop() nice and clean as per Embedded Systems.
- You will not be expected to account for the general state of the room, conditions on the ground, or unexpected conditions in the test area. For example, if your robot snags on the carpet or furniture.
- This SBA is **open-book**. You may make use of any reference to help you **debug** this task.
- A viva voce examination will be conducted at the end of the demonstration. Failure to demonstrate your understanding and involvement in the code will result in an integrity case for further investigation.

**Assessment Task**
You will program your robot to demonstrate the following basic behaviours. The behaviours will be observable and assessed live.

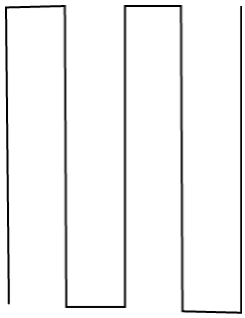
**Square Spiral**
Robot drives in straight lines and turns 90 degrees. Robot may spiral inward, or outward.



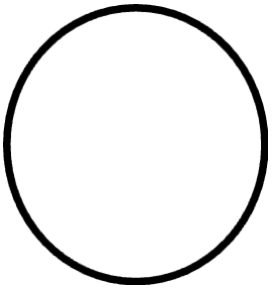
**Zig-zag**
Robot drives in a zig-zag fashion. (Drive-turn-drive-turn-turn-drive, etc)



**Square zig-zag**
Robot drives in a square zig-zag, to cover a square area (similar to mowing a lawn)



**Circle.(Not a spiral)**
Robot drives in a circle. The end point of the circle should meet the start point (Otherwise we're driving in a spiral)



## Marking rubric

### Driving

| Driving Behaviour | Perfect (3 marks) | Mostly on track (2 marks) | Roughly on track (1 mark) | None (0 marks) |
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| **Square Spiral** (3 Marks) | Clean 90° turns, straight lines, even spacing throughout. Sharp corners, no drift. | Recognizable spiral with mostly 90° turns. Minor deviations in straightness or spacing. | Attempts spiral but with significant turn or line issues. Barely recognizable square shape. | No recognizable spiral pattern or basic control. Turns not 90°. |
| **Zig-zag** (3 Marks) | Consistent angles and segment lengths. Smooth, uniform turns. Symmetrical pattern throughout. | Recognizable zig-zag with generally consistent angles. Some variation but clear back-and-forth motion. | Attempts zig-zag but angles vary widely or segments very uneven. Identifiable but poor execution. | Cannot maintain zig-zag pattern. Random wandering, no discernible pattern. |
| **Square Zig-zag** (3 Marks) | Straight parallel lines, precise 90° turns. Even spacing, no gaps/overlaps. Complete coverage. | Recognizable lawn-mowing pattern with parallel lines. Minor issues with straightness or spacing. | Attempts area coverage but lines not parallel, spacing uneven, or areas missed. Intent visible. | Cannot execute area-coverage pattern. Random movement, no systematic structure. |
| **Perfect Circle** (3 Marks) | Smooth continuous circle, constant radius. Endpoint meets startpoint (±5cm). Not a spiral. | Generally circular path that closes well (±10cm). Minor radius variation but clearly a circle, not spiral. | Attempts circle but with flat sections, irregular radius, or poor closure (10-20cm gap). | Cannot maintain circular path. Spirals, very irregular shape, or fails to close. |

### Software

| Task ↓ | Worthy of top marks 9 – 7 marks | Honest attempt to be better than average 6 – 5 marks | Struggling 4 – 3 marks | Insufficient 2 – 1 marks |
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| **Software:** (9 Marks) | • The code is well structured and adheres to recognised best practices.<br>• The methods implemented are efficient, with no unnecessary or nested loops that reduce readability.<br>• Appropriate comments are included to enhance code understanding and maintainability. | • The code is generally acceptable, though it shows occasional lack of organisation; however, a clear effort towards maintaining structure is evident.<br>• Appropriate methods are mostly employed, and the loop() function demonstrates a tidy and logical implementation. | • The code lacks structure and organisation, though it functions to a limited extent.<br>• Most control logic is concentrated within the loop() function, with minimal use of modular methods. | • Code not fit for purpose, does not work.<br>• None or few methods used, used poorly or is non-functional. |
| **Viva voce examination** (9 Marks) | • The response demonstrates full and confident understanding of the program, strongly indicating it was written by the student. | • The response shows clear understanding of the program's structure and logic, indicating the student likely wrote it. | • The response shows limited understanding, with signs that only parts of the code were written or understood by the student. | • The response shows no understanding of the program, suggesting it was not written by the student. |